

Behavioral Finance Research Project

Why Markets Aren't Rational, and What You Can Do About It

A project guide covering research design, methodology, five real-world case studies, the psychology behind each one, the milestone-by-milestone decisions made while building it, the real (not simulated) results obtained, and how the findings translate into practice.

Prepared as a companion guide to the code repository *research/behavioral-finance/*
Written in English for readers without a quant-finance background; every concept is introduced with a plain-language example before it is used.

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Executive Summary

Financial theory has long assumed that markets are, to a first approximation, rational: prices reflect available information, and systematic mispricing should be rare and quickly corrected by sophisticated investors. This project tests that assumption from two directions at once, using real code and, wherever technically possible, real data rather than narrative alone.

Direction one (diagnosis): why do sophisticated, model-driven institutions sometimes fail catastrophically, precisely because their models assumed rational, well-behaved markets? Long-Term Capital Management (1998), the "Quant Quake" (2007), Amaranth Advisors (2006), and the 2008 financial crisis are examined as cases where a locally reasonable model broke because it could not see how fast correlation and panic move together once other market participants stop behaving independently.

Direction two (exploitation): can well-documented behavioral biases — anchoring, underreaction, overreaction — be turned into a measurable, tradable edge, the way real firms such as Fuller & Thaler Asset Management and LSV Asset Management have built businesses on doing? This project builds a signal library and backtest engine, and runs it against a real 21-year daily price history for 48 India-listed companies (the only market data source technically reachable from the environment this project was built in — see Part V for why that constraint mattered and how it was worked around).

The honest headline result from that empirical run: a blended "behavioral score" combining three signals looked like a roughly break-even strategy — and that blend was hiding two very different stories. One component signal (anchoring to the 52-week high) lost money persistently and dramatically; another (short-term reversal) had a real, positive, cost-adjusted edge. The project's central methodological lesson — never trust a blended score without checking what is inside it — came directly out of this finding, not from a textbook.

The guide that follows explains every concept used (behavioral and quantitative) in plain language with real examples, walks through all five case studies in detail, narrates the project's milestones and the reasoning behind each decision, and translates the findings into an investment framework, a risk-management playbook, and a business idea.

Part I — Why This Project Exists

1.1 The theory markets are supposed to follow

The Efficient Market Hypothesis (EMH), the dominant framework in academic finance since the 1960s-70s, holds that asset prices reflect all available information at essentially every moment. Its practical implication is stark: if EMH holds strictly, no strategy should reliably beat a passive index over time, because any predictable pattern would immediately be traded away by rational, profit-seeking investors the moment it appeared.

EMH is not a strawman invented for this project to knock down — it is a genuinely useful default. Most of the time, for most large, liquid securities, prices really do incorporate public information quickly. The interesting question is not "is EMH ever true" (it often is, approximately) but "when and how does it fail, and can those failures be anticipated, measured, or profited from?"

1.2 Two stories that motivate the whole project

TWO EPISODES THAT FRAME THE PROJECT

In 1998, a hedge fund whose partners included two Nobel-laureate economists (Myron Scholes and Robert Merton, co-creators of the option-pricing theory taught in every finance program) lost about 90% of its capital in roughly six weeks, despite running trades its own models considered close to riskless. In January 2021, a struggling video-game retailer's stock rose by roughly an order of magnitude in a few weeks, not because of any change in the company's prospects, but because a large, loosely coordinated group of retail traders on a Reddit forum decided to buy it — and forced some of the most sophisticated short-selling hedge funds in the world to take large losses covering their bets. Both episodes are covered in full in Part IV.

These two stories point in opposite directions — one is about a sophisticated model failing, the other about an unsophisticated crowd winning — but they share a single underlying fact: humans, not equations, decide what a price is. A model can be mathematically elegant and still miss the fact that, under stress, people (whether panicking institutional traders or excited retail traders) start behaving in a correlated, non-independent way that no amount of historical calibration prepared it for.

1.3 The project's two-track design

Two concrete, falsifiable questions structure everything that follows:

- **Q1 (diagnosis).** Can the mechanism by which rational-market risk models fail under stress be shown mechanically, with real numbers, rather than just asserted narratively?
- **Q2 (exploitation).** Do specific, named behavioral biases produce a real, measurable, cost-adjusted return premium when turned into a systematic trading signal, over a long sample, on real market data?

DECISION & RATIONALE

The project could have been written as a pure literature review — summarizing what Kahneman, Tversky, Thaler, and others found, and describing LTCM and GameStop narratively. That would have been faster, but it would not have produced anything falsifiable: a narrative account can always be told to confirm whatever conclusion the writer wanted. The decision here was to require runnable code and, wherever technically possible, real executed numbers for every claim — even when (as happened repeatedly, see Part V) the environment made that harder than expected. A claim in this project is either backed by a number that was actually computed, or it is explicitly flagged as not yet validated. Nothing in between.

Part II — Core Behavioral Concepts, Explained Before They're Used

Every case study and every trading signal in this project rests on one of the concepts below. Each is introduced with an everyday example first, then the specific market mechanism it produces, so that later sections can use the term without re-explaining it.

2.1 Anchoring

Everyday example: if a car salesperson opens negotiation at \$30,000, most buyers end up settling closer to that number than if the same car had been opened at \$25,000 — even though the opening number carries no real information about the car's actual worth. The first number seen becomes a psychological reference point (an "anchor") that subsequent judgments are adjusted from, insufficiently.

Market mechanism: investors use a stock's 52-week high as exactly this kind of reference point. Even when good news arrives that should push the price well past the old high, many investors hesitate to buy through a level they've mentally anchored on as "expensive," so the price drifts up slowly instead of jumping — the mechanism behind the 52-week-high signal used in this project's Q2 signal library (Part III) and central to the George & Hwang (2004) academic finding it's based on.

2.2 Herding (Social Proof)

Everyday example: seeing a long line outside one restaurant and an empty one next door, most passersby join the line — not because they have independently evaluated either restaurant's food, but because a crowd is itself treated as evidence of quality. This is useful most of the time (crowds often do know something) and can be badly wrong when the crowd itself started for an unrelated reason.

Market mechanism: in a crisis, investors and institutions sell what everyone else is selling, and buy what's flowing toward safety (a "flight to quality"), regardless of the specific fundamental merits of what they're selling or buying. This is the central mechanism in the LTCM case (Part IV): LTCM's own trades were not herding, but the rest of the market's correlated flight to safety broke the correlation assumptions LTCM's model depended on.

2.3 Overconfidence and Illusion of Control

Everyday example: most drivers rate themselves as "above average" — a statistical impossibility for a majority to be true, and a well-replicated psychological finding. A string of good outcomes (safe drives, in this analogy) is easy to misread as evidence of skill rather than of favorable conditions that happened to hold.

Market mechanism: a trader with a strong track record under one market regime can mistake that track record for proof their model handles all regimes, and size positions accordingly. This is the central mechanism in the Amaranth Advisors case (Part IV): trader Brian Hunter's prior profitable natural-gas positions preceded a concentrated bet sized well beyond what the fund could survive being wrong about.

2.4 Underreaction and Overreaction

Everyday example: told a piece of surprising news, some people barely update their prior opinion at all (underreaction — "I'll believe it when I see more evidence"), while others swing to the opposite extreme immediately (overreaction — "this changes everything"). Both are documented, opposite failures of the same rational-updating standard (Bayesian updating: revising a belief by exactly as much as new evidence warrants, no more, no less).

Market mechanism, underreaction: after a company reports a surprising earnings beat, its stock price often keeps drifting upward for weeks afterward instead of jumping immediately to the new fair value — a well-documented anomaly called post-earnings-announcement drift (PEAD), because analysts and investors update their models too slowly. This underreaction is also the mechanism behind price *momentum* more generally: information diffuses slowly, so a trend already underway tends to continue.

Market mechanism, overreaction: a sharp, sudden price move (a panic sale or a euphoric rally) often partially reverses over the following weeks as the overreaction unwinds — the mechanism behind the short-term *reversal* signal used in this project, and, as it turned out, the single signal that actually worked in the real backtest (Part V).

2.5 Extrapolation Bias

Everyday example: a company that has grown fast for three years straight is often assumed, informally, to keep growing at a similar pace — even though few businesses sustain any given growth rate indefinitely. People extrapolate recent trends further into the future than the trends themselves justify.

Market mechanism: investors chase exciting recent growth stories ("glamour" stocks) and shun boring, currently out-of-favor companies, bidding growth stocks up beyond what their fundamentals justify and value stocks down below theirs. Lakonishok, Shleifer & Vishny's 1994 paper (the founding research behind LSV Asset Management, Part IV) argued this extrapolation bias, not extra risk, explains much of the historical outperformance of value stocks.

2.6 Attention and Availability Bias

Everyday example: people tend to overestimate the probability of dying in a plane crash relative to a car crash, because plane crashes are rare but vivid and heavily covered by media, so examples come to mind (are "available") more easily than the much more common, less-covered car crash.

Market mechanism: retail investors disproportionately buy stocks that have recently grabbed their attention — extreme returns, heavy news or social-media coverage — rather than researching broadly across the market first (Barber & Odean, 2008). This is the central mechanism behind the GameStop episode (Part IV): a stock became a coordinated retail target largely because it was already attention-grabbing, not because of a broad, independent fundamental re-evaluation by thousands of separate investors.

2.7 Crowding: Herding at Institutional Scale

Everyday example: if every restaurant critic in a city independently decides "fusion cuisine is the next big trend" and reviews accordingly, the city can end up with far more fusion restaurants than genuine demand supports — not because any critic copied another, but because they were all reasoning from similar assumptions and similar data.

Market mechanism: many quantitative hedge funds, working independently with different proprietary signals, can still end up holding very similar positions if their signals are built on the same underlying academic factors (value, momentum). Their portfolios look diversified from the inside but are secretly correlated with each other. This is the central mechanism behind the 2007 "Quant Quake" (Part IV): once one large fund was forced to sell, the selling pressure looked, to every other similarly-positioned fund's model, like a signal to sell the same names.

Part III — Quantitative Methodology, Explained

3.1 What is Value-at-Risk, and why does a "calm-regime" model miss the point?

Plain-language definition: Value-at-Risk (VaR) at the 99% confidence level answers the question "on a bad-but-not-extreme day (one of the worst 1% of days), roughly how much could this portfolio lose?" It is the financial -industry equivalent of an insurer asking "how much should we expect to pay out in a bad year, one we'd expect to see about once every hundred years?"

Why a Gaussian (bell-curve) model of VaR is dangerous: the easiest way to estimate VaR is to assume returns follow a normal (bell-curve) distribution, calibrated on however many years of historical data are available, and to assume the **correlations** between different positions — how much they tend to move together — stay roughly constant. Both assumptions are usually reasonable in ordinary conditions. Both usually fail together, at the worst possible time: during a genuine crisis, returns move by more than a bell curve predicts ("fat tails"), and positions that used to move mostly independently start moving together (correlations spike toward 1), because everyone is reacting to the same underlying fear at once.

ANALOGY: TWO HOUSES AND A WILDFIRE

Two houses on the same street, insured separately against fire. Ordinarily, one house catching fire tells an insurer almost nothing about whether the neighboring house will too — fires are close to independent events, so insuring both together looks safe. But a wildfire changes that: it can burn down both houses at once, because now the two risks share a single common cause. A risk model calibrated only on ordinary house fires will badly underestimate the insurer's exposure to a wildfire year, not because the model is bad at estimating ordinary fires, but because it never saw the correlation-raising event in its calibration data. This is exactly the mechanism this project's Part III.3 simulation quantifies for a leveraged trading book.

3.2 The Q1 risk simulation: setup

Rather than simply asserting that Gaussian VaR models understate tail risk, this project built a Monte Carlo simulation (*risk_simulation/fat_tails_vs_normal.py* in the code repository) that computes the underestimation directly. It models a leveraged, six-asset relative-value book (weights summing to roughly 9x gross exposure, loosely evocative of the kind of leveraged spread trades LTCM ran) under two return-generating processes:

- **The model the risk desk believes:** a Gaussian (bell-curve) distribution, with a modest, constant 0.25 pairwise correlation between assets, calibrated on "calm regime" data — exactly the kind of risk model a desk would build from a few years of ordinary historical returns.
- **The process that actually generates the data:** the same calm regime most days, but with a small (1%) daily probability of switching into a "stress regime" where volatility roughly triples and pairwise correlation jumps to 0.92 — combined with fat-tailed (Student-t) shocks in both regimes for realistic extreme-day behavior.

DECISION & RATIONALE

This project's sandbox environment had no general internet access when this simulation was built (see Part V for the full story), which ruled out downloading real historical crisis data to measure this effect directly. Rather than skip the question or fake a result, the decision was to build a Monte Carlo simulation that needs no external data at all — every number it reports is a genuine, reproducible output of code that actually ran, even though the underlying scenario is stylized rather than a literal reconstruction of any single historical event. The parameters (0.25 to 0.92 correlation, 3x volatility) were chosen to be qualitatively consistent with what is publicly known about the LTCM episode's shape, not fit to LTCM's actual (never fully disclosed) book — the report is explicit that the resulting multiple should be read as "this is the order of magnitude of the effect," not a precise historical reconstruction.

3.3 The Q1 results (real, executed numbers)

Tail depth	Gaussian (calm-regime) VaR	True (simulated) VaR	True Expected Shortfall (CVaR)
99.0%	7.67% of book	8.15% of book (1.06x)	10.37% of book (1.35x)
99.9%	10.19% of book	13.26% of book (1.30x)	16.83% of book (1.65x)

Worst single day across 500,000 simulated trading days: a loss of 58.9% of the book — a scale of move a Gaussian, calm-regime model treats as essentially impossible, but which falls naturally out of a process that occasionally lets volatility and correlation rise together. The gap between the Gaussian model and reality is modest right at the 99% line (that quantile happens to sit near the boundary between regimes in this parameterization) but compounds quickly deeper into the tail — which is itself an important lesson: a risk model can look adequate at the confidence level a regulator or risk committee checks routinely, and still be dangerously wrong exactly where it matters most, a few notches further into the tail.

3.4 Turning behavioral biases into tradable formulas

Part II described three biases in plain language. Here is how each becomes a number that can be computed for every stock, every day, and used to rank a universe of companies:

Signal	Formula (plain language)	Bias it targets
12-1 month momentum	Cumulative return over the trailing ~12 months, excluding the most recent month (to avoid contaminating the trend)	Underreaction
52-week-high proximity	Current price divided by the highest price reached in the trailing 52 weeks (1.0 = at the high)	Anchoring
Short-term reversal	Negative of the trailing ~1-month return (a recent loser scores positively, betting on a bounce)	Overreaction

A **composite score** blends all three by converting each to a cross-sectional z-score (how many standard deviations a stock is above or below the average stock, on that signal, on that day) and averaging — the intended purpose being a single number that combines all three biases into one rank. Part V explains why this turned out to be a more dangerous simplification than it first appears.

3.5 How the backtest actually works

The methodology follows the standard approach used in the academic momentum and reversal literature, so results are comparable to published findings rather than being an artifact of a bespoke design:

- Once a month, every stock in the universe is ranked by its current signal score.
- The strategy goes equally-weighted *long* the top group (the best-ranked names) and equally-weighted *short* the bottom group (the worst-ranked names) — a "long-short" portfolio, so its return reflects the spread between the best and worst names, not the market's overall direction.
- The portfolio is held for one month, then re-ranked and rebalanced.

Three metrics summarize the result:

- **Sharpe ratio:** average return divided by the volatility (bumpiness) of that return, annualized. A Sharpe of 1.0 is generally considered good for a systematic strategy; a Sharpe near 0 means the strategy earned close to nothing per unit of risk taken; a negative Sharpe means it lost money relative to its own risk.
- **Maximum drawdown:** the largest peak-to-trough decline the strategy experienced at any point in the backtest — a measure of the worst single stretch an investor in the strategy would have had to sit through.
- **Turnover and transaction costs:** every rebalance that changes which names are held costs money in practice (bid-ask spread, market impact). The backtest engine tracks how much of the portfolio changes each month (turnover) and deducts a simple cost (basis points per unit of turnover) to report both a "gross" return (before costs) and a "net" return (after costs) — because a strategy that looks good gross and disappears net of realistic costs is a very common way quantitative research misleads itself.

Part IV — Five Case Studies, in Depth

Each case below follows the same structure: what happened, with a timeline; which behavioral mechanism from Part II explains it, in plain language; and why the case matters for this project's central question.

4.1 Long-Term Capital Management (1998)

What happened

Long-Term Capital Management was a hedge fund founded in 1994 by John Meriwether, with partners that included Myron Scholes and Robert Merton, who shared the 1997 Nobel Memorial Prize in Economic Sciences for the Black-Scholes-Merton option-pricing framework taught in every finance curriculum. The fund ran highly leveraged convergence/relative-value arbitrage: betting that prices of closely related securities would converge toward their historically "fair" relationship. Individual trades had small expected edges, so the fund used enormous leverage — over 25:1 on its balance sheet, with derivative notional exposure exceeding \$1 trillion — to turn small, statistically reliable-looking edges into a large return.

From 1994-1997 the fund returned roughly 20-40% annually, a track record that built exactly the kind of confidence Part II.3 (overconfidence) describes. Russia's default and rouble devaluation on August 17, 1998 triggered a global "flight to quality": investors worldwide fled toward the safest, most liquid assets (US Treasuries) and away from everything else, regardless of the specific fundamental link between the assets they were selling. Spreads LTCM's model treated as largely independent instead moved together, in the same direction, and kept widening instead of converging. The fund lost about \$1.9 billion in August alone; by September 21, capital had fallen from roughly \$4.7 billion at the start of the year to about \$600 million, against essentially unchanged notional exposure. On September 23, 1998, with the fund days from a disorderly default that regulators feared could force a fire-sale across Wall Street, the Federal Reserve Bank of New York organized (but did not fund) a \$3.625 billion recapitalization by 14 banks in exchange for 90% of the fund's equity.

The behavioral mechanism, explained

This is a herding case (Part II.2), but with an important twist: the herding was not LTCM's own. LTCM's individual trades were, in isolation, defensible. What broke the model was that every other market participant started behaving in a correlated way at once — a collective flight to safety that a "rational agents, independent decisions" model does not naturally produce. The lesson generalizes: a model of rational markets is only as good as its assumption that other participants keep behaving independently under stress, and that is precisely the assumption that breaks in a genuine crisis — the exact mechanism quantified in Part III's Q1 simulation.

Why it matters here

LTCM is the founding case for this entire project: a fund staffed by the people who literally wrote the mathematics of modern option pricing still lost 90% of its capital in six weeks, not because the mathematics was wrong, but because the mathematics assumed away exactly the kind of correlated human behavior that shows up under real stress.

4.2 The "Quant Quake" (August 2007) and Amaranth Advisors (2006)

What happened — Quant Quake

Between roughly August 6-9, 2007, a large number of market-neutral quantitative equity funds — using different individual signals, run by different firms, with no apparent common trigger in company fundamentals — suffered simultaneous, sharply correlated losses over a few trading days, followed by a partial rebound once the selling pressure eased. Several well-known quant strategies lost on the order of 5-10%+ in that single week — an extreme move for supposedly market-neutral, low-volatility books. The standard academic reconstruction (Khandani & Lo, 2007/2011) attributes this to **crowding**: many quant managers had, independently, arrived at similar factor tilts (value, momentum), so their books were far more correlated with each other than any of them individually realized. When one or more large funds were forced to deleverage a quant book quickly (for reasons believed unrelated to those specific positions), the forced selling looked, to every other similarly-built model, like a signal to sell the same names.

What happened — Amaranth

Amaranth Advisors was a large multi-strategy hedge fund that lost approximately \$6.5 billion in September 2006, almost entirely from concentrated, highly leveraged natural-gas futures and options positions run by a single trader, Brian Hunter. Hunter's core position was a calendar spread — long winter-month natural-gas futures against short summer-month contracts, betting the spread would widen — sized large enough, relative to actual market open interest, that Amaranth's own trading could move the market it was betting on. A mild start to the 2006 hurricane season pushed the spread the wrong way; over roughly a week in mid-September 2006 the fund lost more than half its capital.

The behavioral mechanisms, explained

Quant Quake is a crowding case (Part II.7): herding among the most sophisticated participants in the market, the group a naive efficient-markets story would expect to be immune to it. Amaranth is an overconfidence case (Part II.3): Hunter's prior profitable natural-gas trades made a concentrated, survivable-only-if-right position look more justified than the fund's risk limits should have allowed.

Why they matter here

Together, these two cases show that neither sophistication (Quant Quake) nor a genuinely skilled track record (Amaranth) protects against the specific behavioral failure modes this project studies — they just change which failure mode shows up.

4.3 The 2008 Financial Crisis and the Gaussian Copula

What happened

The 2007-2008 crisis is usually told as a subprime mortgage story, but the mechanism that turned a real but containable housing problem into a systemic banking crisis mirrors this project's Q1 simulation almost exactly. Banks packaged mortgage pools into collateralized debt obligations (CDOs), tranching by seniority. A senior tranche's safety depends critically on how correlated the underlying mortgage defaults are: independent defaults make a senior tranche very safe; correlated defaults (a national downturn hitting every region at once) make that protection evaporate. The industry-standard tool for pricing this correlation risk was the Gaussian copula (David X. Li, 2000), which reduced the hard problem of modeling thousands of correlated defaults to a single correlation parameter, estimated from a history drawn almost entirely from a multi-decade housing boom during which national home prices had never fallen.

When US home prices began falling nationally in 2006-2007 — a regime the calibration window had essentially no examples of — mortgage defaults stopped being independent and started moving together, for the same underlying reason, in every region at once. Senior tranches priced as extremely safe took losses far larger and faster than the model implied, and because these instruments were held throughout the banking system, the losses hit systemically important institutions almost simultaneously (Bear Stearns, March 2008; Lehman Brothers' bankruptcy, September 15, 2008; AIG's roughly \$182 billion federal rescue, driven largely by credit default swaps written against these same instruments).

The behavioral mechanism, explained

This is structurally the identical mistake as a calm-regime Gaussian VaR model (Part III.1's wildfire analogy): a correlation parameter estimated from a period that never included the correlation-raising regime, and therefore blind to how fast that correlation could rise once it arrived. Two behavioral/institutional layers sit on top of the pure model failure: **herding into a shared industry model** (using it was a competitive necessity even for desks that suspected it understated risk — a rational individual response to career/franchise risk producing an irrational collective outcome), and a **correlated trust collapse** once losses started (nobody could tell who was exposed to how much bad paper, so short-term lenders pulled back from everyone at once — the same flight-to-quality mechanism as LTCM, at systemic scale).

Why it matters here

This is the largest-scale version, in modern financial history, of the pattern this project's Q1 simulation makes explicit in miniature: a model that is locally reasonable, calibrated on real data, and wrong specifically about how correlation behaves in the rare regime that does the damage.

4.4 GameStop and the "Meme Stock" Squeeze (January 2021)

What happened

Through 2020, several large hedge funds, most prominently Melvin Capital, held large short positions in GameStop (GME), a struggling brick-and-mortar video-game retailer, on a conventional view that the business was in structural decline. Starting in mid-to-late January 2021, a large, coordinated wave of retail buying — organized largely on the Reddit forum r/wallstreetbets — pushed the share price up by roughly an order of magnitude in a few weeks. Two mechanisms compounded pure buying pressure: a **short squeeze** (as the price rose, short sellers faced margin calls and were forced to buy shares to close losing positions, pushing the price up further) and a **gamma squeeze** (heavy buying of short-dated call options forced the market-makers who sold those calls to buy the underlying stock to stay hedged, an amplifying loop running through market structure rather than anyone's fundamental view). Melvin Capital's short position lost the fund a very large share of its capital in January 2021 and received a \$2.75 billion capital injection from Citadel and Point72 on January 25, 2021. Robinhood and other brokerages restricted buying of GameStop and similar stocks on January 28, 2021, citing clearinghouse deposit requirements — a decision that triggered US congressional hearings in February 2021.

The behavioral mechanisms, explained

This case runs the *opposite* direction from the others: retail herding (Part II.2, at internet speed and scale) overwhelming institutional positioning, amplified by attention-driven trading (Part II.6 — GameStop became a target largely because it was already attention-grabbing, consistent with Barber & Odean's research on individual investors disproportionately buying stocks that recently grabbed their attention) and by the gamified design of mobile-first trading apps.

Why it matters here

GameStop is included specifically to prevent this project's thesis from collapsing into "institutions are rational, retail is irrational" — a genuinely coordinated crowd can overwhelm sophisticated positioning just as forcefully as institutional herding overwhelmed LTCM's models, just pointed in the opposite direction.

4.5 The Other Side: Funds Built to Exploit These Biases

Every case above is a story about a bias causing a loss. These three firms are the counter-evidence: specific, named biases treated as a persistent, exploitable source of return.

Fuller & Thaler Asset Management

Founded in 1993 by Richard Thaler — awarded the 2017 Nobel Memorial Prize in Economic Sciences for work on how limited rationality affects economic decisions — together with Russell Fuller. Their signature mechanism is closely related to Part II.4's underreaction concept: post-earnings-announcement drift, the well-documented finding that stock prices keep drifting in the direction of an earnings surprise for weeks after the announcement, because analysts anchor on their pre-announcement estimates and update too slowly. Their flagship retail vehicle has long traded as "Undiscovered Managers Behavioral Value" (ticker UBVLX).

LSV Asset Management

Founded in 1994 by academic economists Josef Lakonishok, Andrei Shleifer, and Robert Vishny, whose own research (Lakonishok, Shleifer & Vishny, *Journal of Finance* 49(5), 1994, pp. 1541-1578) argued that much of value stocks' historical outperformance over "glamour" growth stocks is explained by investors extrapolating recent growth too far into the future (Part II.5), not by value stocks simply being riskier.

AQR Capital Management

Founded in 1998 by Cliff Asness, whose University of Chicago PhD dissertation (under advisors including Eugene Fama, himself closely associated with the efficient markets hypothesis) was on momentum in stock returns. AQR treats behavioral anomalies as one input among several (alongside risk-based explanations), run with heavy quantitative discipline — a useful middle case between Fuller & Thaler's purely behavioral framing and a pure risk-based factor story.

The working assumption behind all three — and behind this project's signal library in Part III — is that specific, named biases produce persistent, not merely historical, mispricing. Part V.2's empirical result tests that assumption directly, with a real, non-cherry-picked outcome.

Part V — Project Journal: Milestones, Decisions, Results

This part narrates the project as it actually happened, in order, including the constraints that forced decisions and the results that changed the project's direction. Every decision below is stated together with the reasoning behind it, per this guide's brief: nothing is presented as self-evidently correct without the "why."

5.1 Scoping the project

The project started from a broad brief: research the role of human irrationality in markets, with concrete applications to finance and business. Before writing anything, the scope was narrowed by asking four questions: what output format was wanted, which behavioral angle to anchor on, what methodology/evidence to use, and what concrete impact the research should produce.

DECISION & RATIONALE

The answers chosen were: a **quantitative research project with runnable code** (not a pure literature review); anchored on **both** failed "rational market" bets (LTCM-style) **and** behavioral funds that exploit bias; using **public market data plus case-study comparison**; aimed at producing an **investment framework, risk-management lessons, and a business idea**. This combination was chosen specifically because it forces the project to be falsifiable in both directions at once (Part I.3) rather than only telling the flattering half of the story (either "markets are irrational, here's how to profit" or "here's why smart people fail," without testing either claim against real numbers).

5.2 Milestone 1 — Building the pipeline under a hard constraint

The first concrete obstacle appeared immediately: the development environment's outbound network access is restricted by an egress proxy to a specific allowlist. Direct testing (curl requests to each host, and reading the proxy's own status endpoint, which logs connection failures) confirmed that Yahoo Finance, Stooq, SEC EDGAR, and FRED — every conventional free market-data source — were all blocked with a 403 policy denial.

DECISION & RATIONALE

The tempting shortcut here would have been to generate synthetic "market-looking" data and present backtest results computed on it as if they were real findings — exactly the kind of model-dressed-as-reality mistake this project's whole thesis warns against. The decision instead was to build the complete, real pipeline (data loader, signal library, backtest engine) as production-quality code, but to sharply separate two categories of claim: results that were **actually computed by running code in this environment** (which get stated as fact) versus code that is **complete and believed correct but not yet exercised on real data** (which gets stated as an explicit, flagged limitation, never as a finding). The one part of the research that needed no external data at all — the Q1 tail-risk simulation (Part III.2-3.3) — was built first specifically because it could produce a genuine, real result despite the network constraint.

Milestone 1 shipped: the research design document, the full signal and backtest codebase (validated only against synthetic test fixtures at this point, with that limitation stated explicitly), the Q1 risk simulation with its real executed numbers, three initial case studies (LTCM, Quant Quake / Amaranth, the behavioral funds), and a first version of the practical framework document.

PROCESS DECISION

At the end of milestone 1, a new working practice was established at the user's request: after every milestone, provide a recap of the reasoning behind what was built, followed by explicit questions checking whether the project's focus was still correctly calibrated to what was actually wanted. This is why milestones 2 and 3 below each open with what the previous recap's questions surfaced.

5.3 Milestone 2 — Real data, and a result that changed the project

The milestone-1 recap asked, among other things, how to handle the no-internet constraint going forward. The chosen answer was to actively search for a reachable data source rather than accept the limitation as final.

DECISION & RATIONALE

Rather than assume every external host was blocked, a range of unrelated hosts was tested directly — not just financial-data providers but also generic ones (a plain package index, a generic search engine). This showed the block was not "no internet at all" but a specific allowlist: *api.github.com* and *raw.githubusercontent.com* were reachable while general financial-data hosts and even a plain general-purpose website were not. That distinction mattered: it meant GitHub-hosted datasets were a real option, not a dead end.

A web search for GitHub-hosted multi-ticker daily price datasets surfaced a real candidate: a community-uploaded CSV of 63 raw ticker symbols covering NSE (India) large/mid-cap companies from 2000 to 2021. Before using it, the data was checked for basic quality (no duplicate date/symbol rows, no zero or negative prices) and then for a subtler issue: several apparent "different companies" turned out,

on inspection of their trading date ranges, to be the *same* company under a former ticker symbol.

DECISION & RATIONALE

The evidence for this was date-range contiguity: symbol A's last trading date and symbol B's first trading date lined up with no overlap and no gap, for 14 separate symbol pairs (or longer chains) — the exact signature of an NSE ticker rename. Each chain was then cross-checked against publicly known Indian corporate history (for example, "TELCO" was Tata Motors' ticker before a rename; "SESAGOA" became "SSLT" then "VEDL" as Sesa Goa became Sesa Sterlite became Vedanta Ltd) before merging. One pair, HDFC and HDFCBANK, was deliberately *not* merged, because the two were genuinely separate listed companies for the entire sample period (they only merged in 2023, after the data ends). This is the kind of judgment call that is easy to get wrong silently in a backtest — merging two unrelated companies, or failing to merge one company's own history — so it was done from evidence, and locked in afterward with a dedicated automated test (visible in the code repository's `tests/test_loaders.py`) rather than left as a one-off manual step.

With 48 clean, continuous company series covering 21 years, the full signal-and-backtest pipeline was run for the first time on real market data.

Signal	Split	Gross annual return	Gross Sharpe	Net Sharpe	Max drawdown
Composite (all 3)	deciles	-3.59%	0.01	-0.03	-74.3%
Composite (all 3)	quintiles	-1.33%	0.05	0.00	-63.2%
12-1 momentum only	quintiles	-1.73%	0.04	0.01	-60.8%
52-week-high only	quintiles	-13.71%	-0.51	-0.54	-97.1%
Short-term reversal only	quintiles	+4.46%	0.30	0.22	-64.1%

The composite score alone would have supported a bland, roughly-correct-sounding conclusion: "no real edge here." Decomposing it into its three parts told a much more informative and more dangerous story: the 52-week-high (anchoring) signal was actively destructive, losing money persistently and suffering a 97% drawdown, while the short-term reversal (overreaction) signal had a genuine, positive, cost-adjusted edge. Averaging them together hid both facts.

A follow-up check tested whether the anchoring signal's damage was simply an artifact of the 2008 and 2020 crashes (both periods where a "buy recent winners" strategy is known from the academic literature to be exposed to sharp reversals — "momentum crash risk"). Excluding both crash windows entirely, the strategy's Sharpe ratio outside them was still -0.46, and its cumulative loss *excluding* the crashes (-90%) was actually larger than its loss *during* them (-53%) — ruling out "it only lost money in the two crashes" as the explanation, and leaving the true cause an open question flagged honestly rather than resolved with a plausible-sounding guess. (Part V.4 below picks this exact question back up.)

DECISION & RATIONALE

The milestone-2 recap surfaced this same lesson back to the practice guiding the project itself: a mediocre-looking blended result should always be decomposed before being trusted, a rule now stated explicitly in the practical framework (Part VI.1). The recap also asked whether tying the eventual business idea to this repository's own existing software platform (a private-equity deal-screening tool the code happened to live alongside) was the right framing. The answer was no — the business idea was rewritten from scratch as a standalone analytics service (Part VI.3), decoupled from any specific existing platform, per that explicit feedback.

5.4 Milestone 3 — Replication, and a rejected hypothesis

The milestone-2 recap's questions produced three directions: search for a second, independent market's data to check whether the NSE findings replicate elsewhere; investigate *why* the 52-week-high signal lost money, specifically testing whether it was confounded with a value/growth effect; and continue expanding the codebase rather than wrapping up. Producing this guide itself was folded into the same milestone, at the user's explicit request to run it in parallel with the quantitative work rather than after it — which is why an earlier draft of this section reported milestone 3 as in-progress; both investigations below have since completed.

Finding a second market

A GitHub-hosted mirror of the well-known Kaggle "Huge Stock Market Dataset" (one CSV per US-listed ticker) was located and verified: 30 US large-cap names, the same tickers as the project's original default universe, each starting at its own listing date and running through 2017-11-10 (the dataset's last snapshot; this is a historical replication check, not a live feed).

Signal	Split	Gross annual return	Gross Sharpe	Net Sharpe	Max drawdown
Composite (all 3)	quintiles	+2.99%	0.25	0.21	-79.6%
12-1 momentum only	quintiles	+5.72%	0.34	0.32	-88.4%
52-week-high only	quintiles	-13.98%	-0.29	-0.32	-99.9%
Short-term reversal only	quintiles	-0.68%	0.13	0.07	-83.0%

The result was not a clean confirmation of the NSE finding — it was more informative than that. **Momentum and reversal flipped which one looked real:** reversal was NSE's edge and the US market's near-zero result; momentum was NSE's near-zero result and the US market's real, positive, cost-adjusted edge (Sharpe 0.32 net). Only one thing held up unchanged across both very different markets and eras: **the 52-week-high signal lost money in both.**

DECISION & RATIONALE

A single-market backtest result is provisional by default. Had this project stopped after milestone 2 (NSE only), it would have reported "momentum is dead here, reversal is the real edge" — the opposite of what the US data alone would have suggested. Neither market alone is the right answer; the honest finding is that anomaly presence is universe-dependent for momentum and reversal, while the 52-week-high signal's damage is not.

Testing the value/growth confound — and rejecting it

The leading candidate explanation for the 52-week-high signal's losses, carried over from milestone 2, was that India's two-decade secular bull market could make "short the names far from their 52-week high" equivalent to "short cheap, beaten-down names" that then mean-revert upward — a value effect masquerading as an anchoring effect. This was tested directly: a simple price-only value proxy (current price divided by its own trailing ~3-year average) was built, its cross-sectional correlation with the raw 52-week-high score was measured, and the 52-week-high signal was then **orthogonalized** against it — a date-by-date cross-sectional regression that mathematically removes the shared value component, leaving only the part of the signal that isn't explained by it.

Market	Correlation (signal, value proxy)	Raw signal net Sharpe	Orthogonalized net Sharpe
NSE (India)	0.63	-0.54	-0.73
US (Kaggle mirror)	0.44	-0.32	-0.49

The two signals genuinely are correlated (0.44-0.63) — a stock near its 52-week high does tend to look "expensive" on this proxy. But removing that shared component made performance **worse**, not better, in both markets. If the value/growth confound had been the real explanation, the "purified" signal should have looked neutral or better once the confounding value component was stripped out; instead it got more negative. **The value/growth confound hypothesis is rejected.**

WHAT THIS DOES AND DOESN'T PROVE

This is a negative result, and it was reported as one rather than quietly dropped or reframed as a success. Combined with the milestone-2 finding that the loss isn't concentrated in the 2008/2020 crash windows either, two of the three candidate explanations for the 52-week-high result are now ruled out by direct test, not just by argument. The remaining candidate — generic momentum-crash risk, a documented property of "long recent winners" strategies in general — is the best-supported explanation left standing, but it has not itself been directly tested (that would mean checking whether the losses cluster specifically in high-volatility periods). The project states this as narrowed, not solved.

Part VI — From Research to Practice

6.1 An investment framework: the Behavioral Mispricing Score

The composite score from Part III.4 is designed to be used as a **tilt or flag layered on top of an existing fundamentals-driven screen**, never as a standalone strategy — the same pattern every fund in Part IV.5 actually follows (they all pair behavioral signals with fundamental discipline, not raw signal-following). A stock that screens well fundamentally *and* sits in the top behavioral decile is a stronger candidate than either signal alone would suggest; a stock flagged as a behavioral "loser" deserves a second look even when its fundamentals look fine.

DECISION & RATIONALE

Rule, stated as a direct consequence of Part V.3's finding, not as generic best practice borrowed from a textbook: **never trust a blended score without checking what each of its components does on its own**. The composite here looked mediocre; decomposing it found one component actively losing money with a 97% drawdown and another with a genuine, positive, cost-adjusted edge. Any score this project's code produces should be reported and validated component-by-component before a blended version of it is trusted for a decision.

6.2 A risk-management playbook, from the Q1 simulation

- **Never calibrate tail risk on a calm-regime correlation matrix alone.** The Part III simulation shows a Gaussian VaR model calibrated on "normal" data understates the true 99.9% tail loss by roughly 1.3x, purely because it cannot see correlations rising toward 1 under stress. Any leveraged strategy needs a stress-regime correlation scenario as a second, explicit check, not just a historical-calibration number.
- **Leverage does not just scale losses — it changes which losses are survivable.** LTCM's underlying spread moves were not physically unprecedented; 25:1+ leverage turned a bad quarter into insolvency. Position and leverage limits should be set against the stress-regime tail estimate, not the calm-regime one.
- **Crowding is a correlation risk invisible from inside your own portfolio.** The 2007 Quant Quake shows a well-diversified-looking book can be secretly correlated with every other fund running a similar signal. A practical mitigant: track how crowded a factor appears to be (e.g., the realized correlation of a strategy's returns to a public factor index) and de-risk when it's high, independent of the strategy's own apparent confidence.
- **A winning streak is not evidence against tail risk.** Amaranth's Brian Hunter had a strong record before the 2006 blowup; recent success under one regime is weak evidence a concentrated position is safe under a different one. Concentration limits need an explicit override that doesn't relax just because a book has been working.

- **Treat "the model says it's fine" as a hypothesis, not a fact** — especially near known regime-change triggers (sovereign defaults, liquidity crunches, crowded-factor unwinds). This is the project's central thesis applied reflexively to its own tooling, not just to the markets it studies.

6.3 A standalone business idea: decomposed behavioral signal analytics

Off-the-shelf factor data (momentum, value, quality) is typically sold by large vendors as pre-blended, black-box composites, priced for institutions with large budgets. Smaller systematic funds, family offices, independent RIAs, and research desks either cannot afford that tier, or cannot see *inside* the composite to know which component is actually carrying the edge on their specific universe — exactly the failure mode this project hit directly on the NSE data (Part V.3).

DECISION & RATIONALE

This idea was originally framed as an extension of the specific software platform this research happened to be built alongside. That framing was dropped after explicit feedback that the business idea should stand on its own, independent of any particular existing product. The pitch below reflects that: a standalone service, differentiated on the same transparency principle (decomposed, auditable signals; never a pre-blended black box) that Part V.3's finding demonstrated the value of directly.

The product: a subscription analytics service with two parts — a **signal side** (the momentum / 52-week-high / reversal library, run per client against *their* universe, reported as separate, individually-backtested components, never pre-blended) and a **risk side** (the regime-switching stress-VaR methodology from Part III, run against a client's actual position correlations and leverage, reporting calm-regime vs. stress-regime tail loss side by side).

Target customer: small-to-mid systematic equity funds, family offices, and independent RIAs priced out of institutional factor-data tiers but sophisticated enough to want decomposed, re-validated signals rather than a black box; secondarily, finance graduate programs and CFA/PE prep courses, as a teaching tool for the exact "don't trust the blend" lesson this project surfaced.

Revenue model: a per-seat analytics subscription, tiered by number of tracked universes/portfolios — the same go-to-market as existing quant factor-data vendors, priced and scoped for the segment those vendors serve poorly, differentiated specifically on transparency: every number traces to runnable code and a stated backtest window, not a proprietary methodology.

Moat: not the signals themselves (all public, published research) — the moat is the discipline of per-client, per-universe decomposed validation instead of a generic pre-blended score, which is exactly what a larger vendor selling a standardized product across all clients structurally cannot do cheaply.

Part VII — Limitations and Honesty Statement

Stated together in one place because a project whose entire thesis is "don't trust a model's output just because it looks rigorous" owes the same discipline to its own outputs.

- **Survivorship bias.** Both the NSE mirror and the US mirror located in milestone 3 use a present-day-chosen company list, not point-in-time index membership. A rigorous version needs a survivorship-bias-free universe (e.g., CRSP, or a maintained point-in-time constituents file).
- **Data provenance.** Both the NSE and US mirrors are personal, community-uploaded GitHub repositories, not official exchange or licensed vendor feeds. Results from them are a genuine methodology demonstration on real prices, not investment-grade research, until cross-checked against an official source.
- **Momentum and reversal did not replicate consistently** across the two markets tested (Part V.4) — treat either result, in isolation, as provisional rather than a confirmed anomaly. Only the 52-week-high signal's loss held up in both.
- **The 52-week-high result's cause is narrowed, not confirmed.** Two candidate explanations (crash-window concentration; a value/growth confound) have been directly tested and rejected (Part V.3-V.4). The remaining candidate, generic momentum-crash risk, is the best-supported explanation left, but has not itself been directly tested.
- **Transaction costs are a simple linear model**, not a real market-impact model; a strategy sized for real capital would need a proper implementation-shortfall estimate.
- **No out-of-sample / walk-forward validation is wired up by default.** A real deployment should split into a strict in-sample fit period and out-of-sample test period, and check for performance decay after each anomaly's academic publication date — a well-documented risk for momentum and reversal specifically.
- **The Q1 simulation's parameters are illustrative**, calibrated loosely to the LTCM episode's qualitative shape (correlations that were low/moderate in normal times moving toward 1 in crisis), not fit to LTCM's actual, never fully disclosed book. Its reported multiple should be read as an order of magnitude, not a precise historical reconstruction.
- **The 52-week-high result's cause is not yet confirmed.** Momentum-crash risk and a value/growth confound are both plausible, neither is proven; Part V.4's in-progress investigation exists specifically to test the second explanation.

Conclusions

The Efficient Market Hypothesis is a useful default, not a law of nature. This project's five case studies show it failing in genuinely different ways — a Nobel-laureate-staffed fund undone by other investors' correlated panic; sophisticated quant funds undone by their own hidden crowding; an entire industry's risk model undone by a correlation assumption nobody's calibration data ever tested; a struggling retailer's stock undone (in the institutional short-sellers' favor, this time) by coordinated retail attention — which is itself evidence against any single, tidy story of "why markets are irrational." There isn't one mechanism; there are several, each traceable to a specific, well-documented piece of human psychology.

The project's real empirical result (Part V.3) delivered a finding more useful than a simple confirmation of either extreme view would have been: not "markets are irrational, biases always pay," and not "the anomalies are all arbitrated away, don't bother." Instead: one specific, named bias (anchoring, via the 52-week-high signal) actively lost money on this universe, while a different one (overreaction, via short-term reversal) had a real, cost-adjusted edge — and the two were invisible from inside a single blended composite. That is, in miniature, the entire project's argument: rigor means checking the specific mechanism, not trusting the reassuring-looking aggregate.

The practical output (Part VI) turns that into three concrete artifacts: an investment framework that explicitly forbids trusting a blend without decomposing it; a risk-management playbook built directly from a real simulated result, not a generic checklist; and a business idea whose differentiation *is* the decomposition discipline the research itself needed. Milestone 3 then ran the India finding through the same skepticism the project applies to everything else, and it did not survive unchanged: momentum and reversal turned out to be universe-dependent, not general truths, while the 52-week-high signal's damage held up across two very different markets and eras, and survived a direct test against its most plausible alternative explanation. That is the project working as intended — not every finding needs to be confirmed to be useful; knowing which findings don't generalize is itself the deliverable.

Glossary

Anchoring

Relying too heavily on an initial reference point (e.g. a 52-week high, an opening price) when making subsequent judgments.

Composite score

A single number blending several individual signals into one rank; this project's central finding is that a composite can hide a component that is actively harmful.

Correlation regime shift

A change in how much different assets move together — typically a jump from low/moderate to near-1 correlation during a market crisis.

Crowding

Many independent, sophisticated participants arriving at similar positions without coordinating, producing hidden correlation across supposedly diversified portfolios.

Decile / quintile backtest

A methodology that ranks a universe by a signal, then goes long the best-ranked group and short the worst-ranked group, to isolate the return spread the signal actually predicts.

Efficient Market Hypothesis (EMH)

The theory that asset prices reflect all available information, implying no strategy should reliably beat a passive index.

Extrapolation bias

Assuming a recent trend (e.g. fast earnings growth) will continue further into the future than it typically does.

Fat tails

A return distribution with more extreme outcomes than a normal (bell-curve) distribution predicts.

Herding

Following the crowd's behavior rather than independently evaluating a decision — useful most of the time, dangerous when the crowd itself started moving for an unrelated reason.

Leverage

Borrowed capital or notional derivative exposure used to scale a position beyond what the investor's own capital would allow; it magnifies both gains and losses.

Maximum drawdown

The largest peak-to-trough decline a strategy experiences at any point in a backtest.

Momentum

The tendency of a security's recent trend to continue, typically attributed to underreaction.

Overconfidence

Overestimating the reliability of one's own judgment or model, often reinforced by a recent string of favorable outcomes.

Post-earnings-announcement drift (PEAD)

The tendency of a stock price to keep drifting in the direction of an earnings surprise for weeks after the announcement, rather than repricing immediately.

Reversal

The tendency of a very recent, sharp price move to partially revert, typically attributed to overreaction.

Sharpe ratio

Average return divided by return volatility, annualized; a standard measure of risk-adjusted performance.

Survivorship bias

The distortion introduced by only including, in a historical study, entities that are still around today (e.g. still-listed companies), which tends to overstate historical performance.

Value-at-Risk (VaR)

An estimate of how much a portfolio could lose over a given horizon at a given confidence level (e.g. the worst-case loss on all but the worst 1% of days).

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All code, full case-study write-ups, and the underlying data-loading and backtest engine referenced throughout this guide live in the project's code repository, folder research/behavioral-finance/, which remains the definitive, most up-to-date source as milestone 3 and beyond continue.